

Environmental drivers of *Plasmodium vivax* malaria in Manaus: Modeling the impact of climate and forest loss

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ABSTRACT

This paper aims to analyze the behavior of malaria transmission in the Amazon region based on climatic and environmental changes, such as temperature, precipitation, humidity, deforestation and forest fires, through proposed modifications to the SIR and SEI models, in order to contribute to the study of applications of external effects on the evolution of the disease. The Trajetórias Project, developed by the Synthesis Center on Biodiversity and Ecosystem Services (SinBiose/CNPq) was used as an initial reference for the study.

This work employs a modified SEIRS/SEI methodology, based on the SIR/SEI model developed by Parham and Michael (2010) which takes into account rainfall and temperature, with further modifications.

1. Introduction

Malaria remains one of the most significant vector-borne diseases worldwide, caused by protozoa of the genus *Plasmodium* and transmitted by female mosquitoes of the genus *Anopheles*. Although present in multiple regions, the burden of malaria in the Americas is highly concentrated in the Amazon basin, which spans nine countries and accounts for a substantial proportion of reported cases [1]. The persistence of malaria in this region is closely linked to ecological, climatic, and socioeconomic factors that influence both vector dynamics and human exposure.

Environmental changes, particularly deforestation and forest fires, have been widely associated with the resurgence and spatial redistribution of malaria in the Amazon [2]. The removal of forest canopy alters local microclimates, increasing sunlight exposure and ambient temperature, which favor the development of mosquito larvae and pupae. The primary Amazonian vector, *Anopheles darlingi*, thrives in partially deforested environments, where sunlit water bodies promote larval growth and the availability of food sources such as algae [3, 4]. Forest fragmentation accelerates mosquito development cycles and increases vectorial capacity and survivorship through higher temperature and light availability [4, 5], while forest fires, often used for land clearing, modify habitat structure and may promote subsequent fires through understory drying and fuel accumulation, contributing to long-term ecological changes that influence mosquito populations and disease transmission dynamics [6, 7, 8].

These processes also shape human exposure to malaria. Activities such as logging, agriculture, and infrastructure development concentrate human populations in recently altered environments, often in close proximity to mosquito breeding

sites, raising human-vector contact rates and facilitating transmission, particularly among non-immune migrants [9, 10]. Additionally, land-use changes such as agriculture may contribute to river sedimentation and the formation of artificial breeding sites, further enhancing mosquito proliferation. Climatic variables such as temperature and rainfall, together with human disturbances such as mining, further influence vector abundance, pathogen development, and transmission intensity. Despite the recognized importance of these environmental drivers, integrating them into mathematical models of malaria transmission remains challenging. In particular, distinguishing between their effects on mosquito biology and their impact on human-vector interaction is essential for developing models that are both biologically consistent and identifiable from available data.

In this work, we analyze and develop modifications to a SIR-SEI model for malaria transmission in the Amazon, inspired by the host-vector framework of Parham and Michael (2010) [11] and the Trajetórias Project [12], which is a dataset of environmental epidemiological and epidemiological indicators, among others, for all municipalities of the Brazilian Legal Amazon. Our analysis focuses on the municipality of Manaus, the largest urban centre of the Brazilian Amazon and that concentrates the highest absolute number of malaria cases in the region. Within the municipality, however, the burden is concentrated in the rural zone, where incidence is substantially higher than in the urban zone or in the municipality as a whole [12], and therefore it was decided to use the rural population of Manaus for the development of our model. The original formulation as proposed by Parham and Michael was not capable of explaining the transmission dynamics we observed in Manaus, and with the modifications described below, our model sets out to do so. The observed series of active infectious cases in Manaus over the 2017-2023 period displays pronounced inter- and intra-annual variability, with recurrent seasonal waves whose timing and amplitude change from year to year,

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which motivates the representation of all of the variability of the environmental drivers and, ultimately, the modifications to the baseline model described below. The model incorporates environmental variables such as temperature, rainfall, humidity, deforestation, and fire activity. Climatic factors, including temperature T , precipitation R , and humidity H , are modeled as drivers of mosquito life-history parameters, while land-use variables are introduced as modifiers of transmission intensity. Deforestation and fire activity are not embedded as mechanistic drivers within the epidemiological model; instead, they are treated as environmental variables that describe the context in which transmission occurs and are used, together with T , R , and H , as predictors in the regression of the weekly per-bite transmission probabilities b_1 and b_2 (see Section 2.10). Land-use and fire activity thereby modulate transmission indirectly, through the empirical relation between environmental conditions and the effective transmission coefficients, rather than by directly altering the mechanistic mosquito rates.

Deforestation was obtained from the DETER system, whose daily alerts were aggregated at weekly resolution, with the 4-week total degraded area (clear cut, forest degradation, and burn scar combined) used as the covariate D_f [13]. Fire activity was represented by two complementary sources: the weekly number of fire hotspots detected by INPE, used as the covariate F_c [14], and annual burned-area trends from the MapBiomas project, assigned as a monthly covariate F_t [15]. Daily temperature, precipitation, and humidity were obtained from the Mosqlimate API [16] and included as the covariates T , R , and H . Together, the variables T , R , H , D_f , F_c , F_t form the environmental predictor set relating observed environmental conditions to the weekly transmission probabilities b_1 and b_2 .

This paper is organized as follows: in Section 2 we present the data and the evolution of the SEIRS-SEI model, starting from the baseline delayed SIR-SEI model all the way to the proposed SEIRS-SEI construction; in Section 3 we present the mathematical analysis of the model, regarding the mosquito population dynamics, the basic reproduction number and the sensitivity analysis; in Section 4 we present the found results; which are discussed in Section 5, and we conclude in Section 6.

2. Methodology

2.1. Epidemiological Data

This study focuses on malaria cases caused by *Plasmodium vivax*, which accounts for the majority of malaria infections in Brazil and is the predominant species in the Amazon region [17, 18, 19]. Epidemiological data were obtained from the SIVEP-Malaria system [20], which provides daily records of notified malaria cases for all Brazilian municipalities from 2003 to 2023. The analysis was restricted to the period 2017-2023, for which complete epidemiological and environmental data were available. Figure 1 presents the total number of active infectious cases observed in rural

Manaus over this period. This zone of the municipality was chosen over the urban or total population due to its higher incidence of cases during the periods studied by the Trajetorias Project [12]. The pronounced inter- and intra-annual variability of this series constitutes the central modelling challenge addressed in this work and motivates the modifications to the baseline model described below.

2.2. Population Estimates

As was said before, the municipality of Manaus was selected due to its high incidence of *P. vivax* malaria, particularly in rural areas. However, detailed yearly estimates of rural population were not directly available from census data. To address this limitation, population estimates were derived using data from the Trajetorias Project [12], which provides incidence values aggregated over five-year periods.

The incidence is defined as

$$\text{Inc}(d, m, z, t_1, t_2) = \frac{\text{Cases}(d, m, z, t_1, t_2)}{\text{Pop}(m, z, (t_1 + t_2)/2) \cdot (t_2 - t_1)} \times 10^5, \quad (1)$$

where $\text{Cases}(d, m, z, t_1, t_2)$ denotes the number of reported cases for disease d in municipality m and zone z over the interval $[t_1, t_2]$, and $\text{Pop}(m, z, \cdot)$ represents the population at the midpoint of the interval.

This relationship was inverted in order to estimate a ratio of rural to total population from reported incidence and case counts in both zones. Using this approach, the rural population of Manaus was estimated to represent less than 1% of the total population, ranging between 0.3% and 0.6% of the total population of the municipality, while accounting for a disproportionately large share of malaria cases (approximately 30% to 40% over the study periods).

Yearly total population values for Manaus were obtained via interpolation based on official census data from the Brazilian Institute of Geography and Statistics (IBGE) for the years 2000, 2010, and 2022 [21, 22, 23]. Intermediate yearly values were constructed using linear interpolation, while values beyond 2022 were extrapolated based on observed trends.

It must be noted that the inter-census estimates of the rural population were derived from fixed rural-urban proportions based on the 2010 census. [12]. While this approach ensures internal consistency, it does not account for the marked demographic changes in Manaus over time, particularly the rapid urbanization observed in the region. To mitigate this limitation, the rural proportions used in this study were recalculated by interpolating and extrapolating the available census data, giving an approximate decrease of 2.38% of the rural population per year, providing a more realistic representation of the temporal evolution of the population. This adjustment is relevant because rural areas contribute disproportionately to malaria incidence. Nevertheless, the

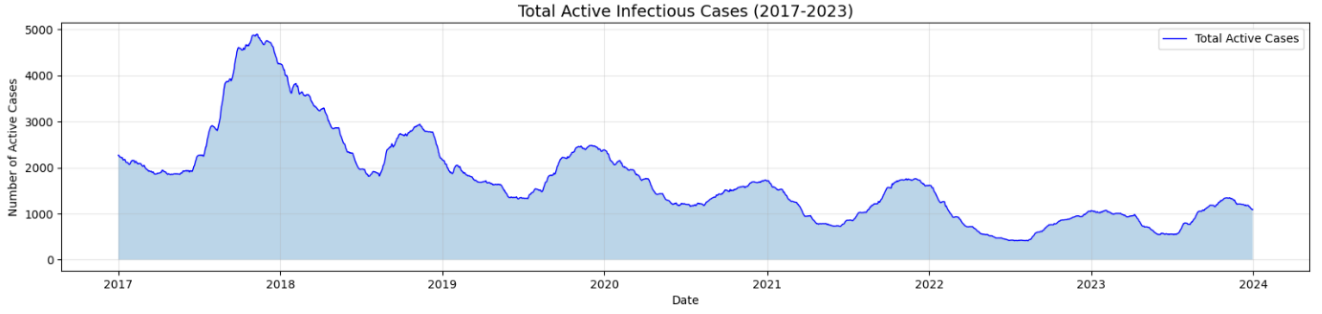


Figure 1: Total number of active infectious cases of malaria (*P. vivax*, SIVEP-Malaria) in rural Manaus, 2017-2023. The strong inter- and intra-annual variability of the disease burden is the central modelling challenge addressed in this work, and motivates the representation of all of the variability of the environmental drivers as well as the modifications to the baseline model.

resulting decrease in N_y during 2017-2023 reflects a demographic trend that may still not correspond to the actual exposure of the human population to mosquitoes.

2.3. Baseline Delayed SIR-SEI Model

We begin the construction of the model with the classical host-vector framework for malaria transmission, based on the model introduced by Parham and Michael (2010) [11], which incorporates temperature- and rainfall-dependent mosquito dynamics. The human population is divided into susceptible, infected, and recovered compartments, denoted by S_H , I_H , and R_H , respectively. The mosquito population is structured into susceptible, exposed, and infectious compartments, denoted by S_M , E_M , and I_M .

The transmission dynamics are governed by the system of delay differential equations:

$$\frac{dS_H}{dt} = -ab_2 \frac{I_M(t)}{N} S_H(t) \quad (2a)$$

$$\frac{dI_H}{dt} = ab_2 \frac{I_M(t - \tau_H)}{N} S_H(t - \tau_H) - \gamma I_H(t) \quad (2b)$$

$$\frac{dR_H}{dt} = \gamma I_H(t) \quad (2c)$$

$$\frac{dS_M}{dt} = b(R, T) - \left(ab_1 \frac{I_H(t)}{N} + \mu(T) \right) S_M(t) \quad (2d)$$

$$\begin{aligned} \frac{dE_M}{dt} = & ab_1 \frac{I_H(t)}{N} S_M(t) - \mu(T) E_M(t) \\ & - ab_1 \frac{I_H(t - \tau_M)}{N} S_M(t - \tau_M) \ell(\tau_M) \end{aligned} \quad (2e)$$

$$\begin{aligned} \frac{dI_M}{dt} = & ab_1 \frac{I_H(t - \tau_M)}{N} S_M(t - \tau_M) \ell(\tau_M) \\ & - \mu(T) I_M(t) \end{aligned} \quad (2f)$$

All parameters are defined in Table A.1, which gives their functional forms and definitions, while state variables are defined in Table A.2, which gives their definitions, units and sources.

Although this formulation incorporates biologically relevant delays, numerical simulations revealed that, under the

parameterization considered, the model fails to reproduce sustained transmission dynamics. In particular, both the infected human population and the mosquito population tends to extinction, indicating that the structure does not adequately capture endemic persistence.

2.4. ODE approximation and structural limitations

To simplify the analysis and avoid the technical complexity of delay differential equations, an ordinary differential equation (ODE) approximation was initially considered. In this formulation, the extrinsic incubation period in mosquitoes is modeled via a transition rate b_3 , replacing the explicit delay τ_M .

The resulting system is given by:

$$\frac{dS_H}{dt} = -ab_2 \frac{I_M}{N} S_H \quad (3a)$$

$$\frac{dI_H}{dt} = ab_2 \frac{I_M}{N} S_H - \gamma I_H \quad (3b)$$

$$\frac{dR_H}{dt} = \gamma I_H \quad (3c)$$

$$\frac{dS_M}{dt} = b(R, T) - \left(ab_1 \frac{I_H}{N} + \mu \right) S_M \quad (3d)$$

$$\frac{dE_M}{dt} = ab_1 \frac{I_H}{N} S_M - (\mu + b_3 \ell) E_M \quad (3e)$$

$$\frac{dI_M}{dt} = b_3 \ell E_M - \mu I_M \quad (3f)$$

While this formulation restores positivity and simplifies numerical implementation, it exhibits a fundamental limitation: the system tends toward a disease-free equilibrium after a single outbreak, failing to capture the recurrent nature of malaria transmission observed in endemic regions. This behavior highlights the importance of explicitly modeling incubation periods and immunity dynamics.

2.5. SEIRS-SEI model with ODE approximation

To address the limitations of the previous formulations, we introduce an ODE based SEIRS-SEI model that incorporates both incubation delays as done above and

temporary immunity in the human population. The human compartments are extended to include an exposed class E_H , and recovered individuals lose immunity at rate ω , returning to the susceptible class.

To account for incubation dynamics in both hosts and vectors, the transition parameters were defined as temperature-dependent development rates.

For humans, the intrinsic incubation process was modeled through $b_{3H} = \frac{1}{\tau_H}$, where τ_H denotes the intrinsic incubation period.

For mosquitoes, the constant transition parameter $b_3 = 1/\tau_M$ from the original formulation was replaced by a Brière-type function,

$$b_{3M}(T) = \eta T(T - T_{\min})(T_{\max} - T)^{1/2}, \quad (4)$$

which incorporates minimum and maximum thermal thresholds for sporogonic development [24, 25].

The scaling coefficient η was estimated by imposing

$$b_{3M}(T_{\text{opt}}) = \frac{1}{\tau_M(T_{\text{opt}})}, \text{ yielding}$$

$$\eta = \frac{\frac{1}{\tau_M(T_{\text{opt}})}}{T_{\text{opt}}(T_{\text{opt}} - T_{\min})(T_{\max} - T_{\text{opt}})^{1/2}} \quad (5)$$

Another addition was the inclusion of a humidity-dependent survival component $p(H)$ on to the daily survival rate of the adult mosquitoes, which is modeled as a sigmoid function of the daily minimum relative humidity H . The parameter H_0 (in %) represents the humidity threshold at which the mosquito experiences a 50% reduction in survival relative to its maximum, and k (in $\%^{-1}$) controls the steepness of the transition. The parameter φ (dimensionless) establishes a baseline survival floor, accounting for the persistence of individuals in favorable micro-habitats even under extreme desiccation conditions.

The SEIRS-SEI complete model is then given by:

$$\frac{dS_H}{dt} = -\lambda_H(t)S_H + \omega R_H \quad (6a)$$

$$\frac{dE_H}{dt} = \lambda_H(t)S_H - b_{3H}E_H \quad (6b)$$

$$\frac{dI_H}{dt} = b_{3H}E_H - \gamma I_H \quad (6c)$$

$$\frac{dR_H}{dt} = \gamma I_H - \omega R_H \quad (6d)$$

$$\frac{dS_M}{dt} = \Lambda(t) - \lambda_M(t)S_M - \mu S_M \quad (6e)$$

$$\frac{dE_M}{dt} = \lambda_M(t)S_M - (\mu + b_{3M}\ell)E_M \quad (6f)$$

$$\frac{dI_M}{dt} = b_{3M}\ell E_M - \mu I_M \quad (6g)$$

The forces of infection are defined as:

$$\lambda_H(t) = a b_2 \frac{I_M(t)}{N(t)} \quad (7a)$$

$$\lambda_M(t) = a b_1 \frac{I_H(t)}{N(t)} \quad (7b)$$

2.6. Recalibration of the temperature-dependent biting rate

In the original formulation by Parham & Michael [11], the biting rate is expressed as a temperature-dependent function derived from the duration of the gonotrophic cycle. Although this formulation captures the general thermal dependence of mosquito feeding behavior, its parameterization was not developed specifically for Amazonian mosquito populations. With this in mind, we recalibrated the biting-rate function using entomological observations reported by de Barros, Honório and Arruda [26], regarding females of *Anopheles darlingi* collected between August 2003 and July 2004 through intermittent sampling carried out in selected months over the study period.

Their results allowed for estimation of the duration of the gonotrophic cycle through the following formula:

$$P = Q^{1/g} \implies g = \frac{\ln(Q)}{\ln(P)}, \quad (8)$$

where P is the daily survival rate of mosquitoes, Q is the proportion of parous mosquitoes females in the population and g is the duration of the gonotrophic cycle, given in days [27].

Given the lack of more recent data, it was decided to use the obtained estimates of the time period specified above with average temperatures from the period of analysis between 2017 and 2023, given that the biting rate is given as a function of temperature. It was decided to consider $T' = 16^\circ\text{C}$, given that below this temperature there is no development of the vector, as the threshold temperature for the biting rate, and we estimated values of g for the missing months of the analysis via interpolation. With this, given the closed formula for the biting rate, and knowing that it is the inverse of the gonotrophic cycle, we calculated $D_1 = g(T - T')$ for each of the months with available data.

Having monthly values of D_1 , we took their average in order to estimate the biting rate and the proportion of female mosquitoes to humans, which is given by

$$m = \frac{HBR}{a(T)}, \quad (9)$$

where HBR is the mean biting density of *An. darlingi*, given in bites/person/night, while $a(T)$ is the biting rate, given in bites/mosquito/night. Based on the data, M_0 , the initial population of mosquitoes used in the analysis, was estimated to be approximately $10N_0$, based on the found value of m for

the month of January.

At this point, we also approximated the ratio of initially exposed to infectious mosquitoes to be their values at the steady state:

$$\begin{aligned} \frac{dI_M}{dt} &= b_{3M} \ell E_M - \mu I_M = 0 \\ \Rightarrow \frac{E(0)}{I(0)} &= \frac{\mu(0,0)}{b_{3M}(0)\ell(0,0)} \approx 3 \end{aligned}$$

This adjustment preserves the functional structure proposed by Parham and Michael while adapting the parameter values to better reflect the feeding behavior of the primary malaria vector in the Amazon region.

2.7. Density-dependent mosquito recruitment and carrying capacity

Mosquito population dynamics were initially modeled using a linear birth–death process of the form

$$\frac{dM}{dt} = b - \mu M, \quad (10)$$

where b represents the birth rate and μ is the per capita mortality rate. However, this formulation does not account for environmental limitations on population growth.

To incorporate the effect of limited environmental resources, we introduce a density-dependent recruitment mechanism inspired by the classical Verhulst logistic model [28],

$$\frac{dM}{dt} = bM \left(1 - \frac{M}{K}\right) - \mu M, \quad (11)$$

where $M(t) = S_M(t) + E_M(t) + I_M(t)$ denotes the total mosquito population, b the per capita recruitment rate, μ the mortality rate, and K the environmental carrying capacity. This formulation captures the reduction in population growth as the mosquito population approaches the environmental limit.

In the present model, this idea is adapted by replacing the constant recruitment term with the density-dependent function

$$\Lambda(t) = b(R, T) K(t) \left(1 - \frac{M(t)}{K(t)}\right), \quad (12)$$

where the environmental dependence is separated into two distinct mechanisms:

1. The climatic modulation of mosquito recruitment through $b(R, T)$, inherited from the original Parham–Michael formulation;
2. The environmental limitation imposed by habitat availability through the carrying capacity $K(t)$.

In this interpretation, $b(R, T)$ governs the rate at which mosquitoes are produced under favorable temperature and rainfall conditions, whereas $K(t)$ determines the maximum

sustainable mosquito population supported by the environment. Consequently, temperature and precipitation affect mosquito dynamics both through reproductive success and through the availability of suitable breeding habitats.

The environmental carrying capacity is defined by

$$K(t) = M' \cdot f_T(T(t)) \cdot f_R(R(t)), \quad (13)$$

where M' denotes the baseline mosquito population size. Given that we require $0 \leq 1 - \frac{M(t)}{K(t)} \leq 1$, we would require M' such that $M' \geq \frac{M}{f_T(T)f_R(R)}$.

The environmental response functions are given by

$$f_T(T) = \min \left(1.0, \frac{T(T - T_{\min A.}) \sqrt{T_{\max A.} - T}}{T_{\text{opA.}}(T_{\text{opA.}} - T_{\min A.}) \sqrt{T_{\max A.} - T_{\text{opA.}}}} \right), \quad (14a)$$

$$f_R(R) = \left(\frac{R}{R_L/2} \right) \exp \left(1 - \frac{R}{R_L/2} \right), \quad (14b)$$

where $f_T(T)$ describes the temperature suitability for mosquito persistence and $f_R(R)$ represents the dual effect of rainfall on habitat availability.

The rainfall response function increases with precipitation under moderate rainfall conditions, reflecting the formation of breeding sites, but decreases under excessive rainfall due to habitat flushing and destruction of immature stages.

Both environmental response functions are normalized so that $0 \leq f_T(T), f_R(R) \leq 1$, ensuring that the carrying capacity remains bounded by the baseline population scale M_0 .

Unlike the classical logistic formulation, where population growth is directly proportional to the current mosquito abundance, the present representation assumes that recruitment is primarily determined by environmental suitability and breeding-site availability. The factor $\left(1 - \frac{M(t)}{K(t)}\right)$ therefore imposes density-dependent regulation by reducing recruitment as the mosquito population approaches the carrying capacity, providing a biologically realistic upper bound on mosquito abundance. The mathematical implications of this formulation are analyzed in Section 3.1.

2.8. Objective function and parameter bounds

With the model structure, data, and environmental predictors defined, we describe the computational procedures used to calibrate the model to the observed case data, including the objective function, the calibration strategy, and the numerical implementation. Model parameters were estimated by minimizing the discrepancy between the simulated infected-human trajectory $I_H(t)$ and the observed active-case counts reported through SIVEP-Malaria, restricted to rural Manaus (scaled by the rural share of reported cases,

41.54%). The objective function is a weighted mean squared error (wMSE):

$$\text{wMSE}(\theta) = \frac{\sum_{i=1}^n w_i (I_H(t_i; \theta) - y_i)^2}{\sum_{i=1}^n w_i}, \quad (15)$$

where y_i denotes the observed active-case count at date t_i , $I_H(t_i; \theta)$ is the model prediction obtained by linear interpolation of the simulated trajectory onto the observation dates, and θ is the vector of calibrated parameters.

The weights w_i follow an exponential decay that prioritizes early-time dynamics within each year:

$$w_i = w_0 \exp(-\lambda d_i), \quad (16)$$

where d_i is the number of days from the start of the year for observation i , $w_0 = 5.0$ is the initial weight, and $\lambda = 0.0015 \text{ day}^{-1}$ is the decay rate. This weighting scheme reflects the greater sensitivity of the early-season trajectory to initial conditions and parameter values, guiding the optimizer toward solutions that correctly capture the onset of the rainy-season transmission peak.

An extinction penalty was added to the objective to prevent the optimizer from collapsing the infected-mosquito population to near-zero levels, which would sever the force-of-infection coupling and render the human transmission parameters structurally unidentifiable:

$$J(\theta) = \text{wMSE}(\theta) + \mathcal{P}(I_M^{\min}(\theta)), \quad (17)$$

where $I_M^{\min}(\theta) = \min_t I_M(t; \theta)$ is the annual minimum of the infected-mosquito compartment, and

$$\mathcal{P}(I_M^{\min}) = \begin{cases} 0 & \text{if } I_M^{\min} \geq I_M^*, \\ B \left(1 + \frac{I_M^* - I_M^{\min}}{I_M^*} \right) & \text{if } I_M^{\min} < I_M^*, \end{cases} \quad (18)$$

with $I_M^* = 10$ (viability threshold) and $B = 10^8$ (penalty base scale). The penalty scales linearly with the depth of extinction, ensuring that partially-extinct trajectories are ranked proportionally rather than receiving a uniform penalty that would flatten the objective landscape.

The nine calibrated parameters were grouped into four biological “topics” for sequential fitting:

1. **Humidity response** (H_0, k, φ): the half-saturation threshold, sigmoid steepness, and baseline survival floor of the humidity-dependent survival function $p(H)$;
2. **Human transmission** (τ_H, γ, ω): the intrinsic incubation period, recovery rate, and immunity waning rate;
3. **Force of infection** (b_1, b_2): the human-to-mosquito and mosquito-to-human transmission probabilities;
4. **Carrying capacity** (M'): the baseline mosquito population scale that determines the environmental carrying capacity $K(t)$.

The parameter bounds are given in Table 1. The lower bound on $H_0 = 20\%$ prevents the humidity response from collapsing (which would drive $\mu \rightarrow \infty$ and extinguish the vector population). The bound on $b_2 \geq 0.01$ ensures a minimum level of mosquito-to-human transmission, while $\tau_H \in [10, 20]$ days constrains the human incubation period to the biologically plausible range for *P. vivax* [29].

Table 1
Parameter bounds for sequential topic calibration.

Parameter	Lower bound	Upper bound
H_0 (%)	20	75
k (% ⁻¹)	0.01	1.0
φ	0.01	0.5
τ_H (days)	10	20
γ (days ⁻¹)	1/150	1.0
ω (days ⁻¹)	1/300	0.1
b_1	0.001	0.5
b_2	0.01	0.5
M'	50,000	$500 \times N_y^{-1}$

¹ N_y denotes the estimated rural population of Manaus for year y .

2.9. Year-by-year sequential calibration

Parameters were calibrated sequentially for each year from 2017 to 2023 using the `lmfit` implementation of differential evolution (DE) [30]. The DE settings were: population multiplier `popsiz` = 6, maximum function evaluations `max_nfev` = 800, convergence tolerance `tol` = 0.01, and random seed fixed at 42 for reproducibility.

Within each year, the four topics were fitted in the fixed order:

$$\underbrace{(H_0, k, \varphi)}_{\text{humidity}} \rightarrow \underbrace{(\tau_H, \gamma, \omega)}_{\text{human}} \rightarrow \underbrace{(b_1, b_2)}_{\text{Fol}} \rightarrow \underbrace{M'}_{\text{capacity}} \quad (19)$$

At each step, only the current topic's parameters were free; all previously fitted parameters were held fixed at their most recent estimates. This sequential strategy reduces the dimensionality of each optimization sub-problem while respecting the biological ordering: the humidity response determines mosquito mortality, which constrains the infection cycle, which in turn determines the force of infection and carrying capacity.

After the sequential pass, a joint refinement was performed using the Nelder-Mead simplex method [31], with all nine parameters free simultaneously, seeded from the sequential solution. A seed-evaluation guard was applied: after each DE run and after each refinement stage, the starting point was explicitly evaluated, and the optimizer's output was accepted only if it improved upon the seed. This prevents the stochastic DE from returning a solution worse than its initialization.

The model was calibrated sequentially across years, with the end-state vector of each year serving as the initial state for the next:

$$\mathbf{z}_0^{(y+1)} = \mathbf{z}_{\text{end}}^{(y)} \quad (20)$$

where $\mathbf{z} = (S_H, E_H, I_H, R_H, S_M, E_M, I_M)$ is the full state vector. The human infected compartment was always reset to the first observed active-case count for the new year:

$$I_{H,0}^{(y+1)} = \text{round}(y_{\text{first}}^{\text{obs}}) \quad (21)$$

while all other compartments were carried forward without modification. Negative end-state values were clamped to zero before carrying.

For the first year (2017), initial-condition fractions were explicitly fitted using Nelder-Mead with 300 function evaluations:

$$S_{H,0} = \alpha_S \cdot N, \quad E_{H,0} = \alpha_E \cdot N, \quad I_{M,0} = \alpha_I \cdot M_0, \quad (22)$$

where $\alpha_S \in [0.3, 0.9]$, $\alpha_E \in [0.001, 0.30]$, and $\alpha_I \in [0.0005, 0.15]$, with the remaining compartments determined by conservation: $R_{H,0} = N - S_{H,0} - E_{H,0} - I_{H,0}$, $M_0 = 10N$, and $S_{M,0} = M_0 - E_{M,0} - I_{M,0}$.

2.10. Weekly beta diagnostic

To evaluate where the climate-driven formulation deviates from what the data demands, a diagnostic calibration was performed in which the effective transmission coefficients were freed as per-week parameters instead of being derived from the biting rate and fixed b_1, b_2 :

$$\lambda_H(t) = \beta_{h,w} \frac{I_M(t)}{N}, \quad \lambda_M(t) = \beta_{m,w} \frac{I_H(t)}{N}, \quad (23)$$

where $w = \lfloor d/7 \rfloor$ is the week index and d is the day number within the year. This yields $2 \times n_{\text{weeks}}$ free parameters per year (typically 106 for a 365-day year), with bounds $\beta_{h,w}, \beta_{m,w} \in [10^{-9}, 3.0]$.

All other model parameters were fixed at the values obtained from the yearly topic calibration. The initial state was reconstructed from the previous year's end-state, with I_H reset to observed data. The optimizer used was `lmfit.minimize` with the trust-region reflective method (`least_squares`), with `max_nfev` = 1200 and `ftol` = 10^{-8} .

The residual vector for the trust-region solver was constructed as:

$$r_i = \sqrt{w_i} (I_H(t_i) - y_i) + \sqrt{\mathcal{P}(I_M^{\min})}, \quad (24)$$

where $\sqrt{\mathcal{P}}$ is injected as a constant offset to preserve the residual vector length while providing a gradient that pushes I_M upward when the mosquito population approaches the viability threshold.

The weekly beta fit serves as a diagnostic baseline: it reveals the best achievable fit when the transmission coefficients are unconstrained by the climate formulation. Persistent deviations between the weekly-optimal $\beta_{h,w}$ and the climate-driven $a(T) \cdot b_2$ indicate periods where the environmental model fails to capture the true transmission dynamics.

There are three main observations that allow us to use the effective transmission coefficients $\beta_{h,w}$ and $\beta_{m,w}$ as per-week free parameters:

(i) The transmission coefficients are intrinsically time-varying: In the mechanistic formulation, the forces of infection are given by $\lambda_H = a(T) b_2 I_M / N$ and $\lambda_M = a(T) b_1 I_H / N$, where the biting rate $a(T)$ is the only environmental factor and the per-bite probabilities b_1 and b_2 are assumed to be constants. This assumption is biologically restrictive because these coefficients summarize several processes that may vary over the transmission season, including parasite development within the vector, vector susceptibility to infection, vector to host contact, and host infectiousness. These processes can be affected by temperature, rainfall, humidity, and changes in vector and host demography [24, 32]. We therefore relax the assumption that b_1 and b_2 remain constant throughout the study period and allow the data to determine how transmission changes over time.

(ii) The weekly fit provides a nonparametric estimate of the latent transmission process: We represent the effective transmission coefficients as latent time-varying functions, $\beta_h(t)$ and $\beta_m(t)$. These functions are approximated as piecewise constant over weekly intervals:

$$\beta_h(t) = \beta_{h,w}, \quad \beta_m(t) = \beta_{m,w}, \quad t \in [7w, 7(w+1))$$

This provides a flexible finite-dimensional representation of possible transmission profiles. The weekly coefficients are estimated by minimizing the weighted residual between the model predictions and the observed malaria cases. Thus, rather than imposing a specific relationship between transmission and environmental covariates, the weekly model provides a flexible, data-driven estimate of the temporal variation in transmission.

The resulting $2 \times n_{\text{weeks}}$ coefficients don't represent independent biological parameters to be estimated, but rather provide a flexible empirical representation of the transmission process. The additional degrees of freedom allow the model to capture temporal changes in transmission that a more constrained model may miss, providing a reference for evaluating the proposed mechanistic model.

(iii) The climate-driven model is nested within the weekly model: The climate-driven model is obtained by imposing the restriction

$$\beta_{h,w} = a(T_w) b_2, \quad \beta_{m,w} = a(T_w) b_1, \quad (25)$$

where b_1 and b_2 are constant over the entire study period. Consequently, the climate-driven formulation is a restricted case of the more flexible weekly model. Because all other model parameters and initial conditions are held fixed, the difference in fit between the two formulations directly measures the extent to which the constant-coefficient assumption limits the model's ability to reproduce the observed cases.

Releasing the coefficients reduces the wMSE by one to two orders of magnitude relative to the climate-driven model. This substantial improvement indicates that the data contain

temporal variation in transmission that cannot be adequately represented by constant b_1 and b_2 .

The weekly estimates are therefore used as a diagnostic benchmark rather than as the final parameterization of the model. They identify the temporal transmission pattern that a parsimonious mechanistic formulation should seek to explain. This approach preserves the distinction between estimating a flexible empirical transmission process and identifying the underlying biological parameters, as the weekly coefficients characterize the observed transmission signal, while the subsequent structural model determines how that signal can be explained by environmental drivers.

Finally, the weekly estimates provide the link between the empirical transmission signal and the environmental model. Since the biting rate $a(T_w)$ is explicitly modeled as a function of temperature, the corresponding weekly per-bite transmission probabilities can be recovered as

$$b_{1,w} = \frac{\beta_{m,w}}{a(T_w)}, \quad b_{2,w} = \frac{\beta_{h,w}}{a(T_w)}. \quad (26)$$

These time-varying estimates represent the per-bite transmission probabilities implied by the observed case dynamics after accounting for the temperature-dependent biting rate. They are subsequently used as empirical targets for regression models relating b_1 and b_2 to environmental drivers. In this way, the weekly fit serves as an intermediate step in the modeling chain

$$\text{environment} \longrightarrow (b_1, b_2) \longrightarrow \text{transmission} \longrightarrow \text{cases},$$

allowing a flexible, data-driven estimate of transmission to be translated into a simplified and mechanistically interpretable environmental model.

2.11. Numerical solver and implementation

All ODE simulations were performed using Python version 3.12.7 (conda-forge distribution), using the LSODA solver as implemented in `scipy.integrate.solve_ivp` (SciPy version 1.17.1), with automatic stiffness detection and order switching. The integration was evaluated on a fine grid of 10 points per day ($\Delta t = 0.1$ day), with the solution interpolated back to daily resolution via linear interpolation. End-state vectors were clamped to zero to prevent negative compartment values from propagating through the year chain.

Climate inputs (temperature, precipitation, minimum humidity) were smoothed using a 14-day rolling mean for mosquito demographic rates (mortality μ , sporogonic development b_{3M} , birth rate Λ), while the biting rate $a(T)$ used raw daily temperature to preserve high-frequency transmission variability.

Data manipulation and analysis were performed using Pandas version 3.0.2, while numerical computations and array operations were carried out with NumPy version 2.4.2.

All computations were performed on a laptop running Windows 11 (64-bit) equipped with a 12th Gen Intel Core i5-1235U processor (10 cores, 12 threads) and 16 GB RAM.

3. Mathematical Analysis of the Model

3.1. Mosquito population dynamics

In this section, we analyze the qualitative impact of the density-dependent recruitment term on mosquito population dynamics. We compare the proposed formulation with the classical Verhulst model and show that the modified structure prevents extinction under suitable environmental conditions.

Proposition 1 (Environmental persistence of the mosquito population). *Assume that $b(R(t), T(t)) \geq b_0 > 0$ and $K(t) \geq K_0 > 0$ for all $t \geq 0$. Then, the mosquito population $M(t)$ governed by*

$$\frac{dM}{dt} = b(R, T) K(t) \left(1 - \frac{M(t)}{K(t)}\right) - \mu M(t) \quad (27)$$

cannot converge to zero as $t \rightarrow \infty$. In particular, $M = 0$ is not an equilibrium and cannot be asymptotically stable.

Proof. We begin by recalling the classical logistic model with mortality:

$$\frac{dM}{dt} = bM \left(1 - \frac{M}{K}\right) - \mu M. \quad (28)$$

In this formulation, the right-hand side vanishes at $M = 0$, so that $M = 0$ is always an equilibrium. Depending on the parameters, this equilibrium may be stable, allowing extinction of the population.

In contrast, in the present model the recruitment term is given by

$$\Lambda(t) = b(R, T) K(t) \left(1 - \frac{M(t)}{K(t)}\right), \quad (29)$$

and the population dynamics satisfy

$$\frac{dM}{dt} = b(R, T) K(t) \left(1 - \frac{M(t)}{K(t)}\right) - \mu M(t). \quad (30)$$

Evaluating the right-hand side at $M = 0$, we obtain

$$\left. \frac{dM}{dt} \right|_{M=0} = b(R, T) K(t). \quad (31)$$

By assumption, $b(R(t), T(t)) \geq b_0 > 0$ and $K(t) \geq K_0 > 0$, hence

$$\left. \frac{dM}{dt} \right|_{M=0} \geq b_0 K_0 > 0. \quad (32)$$

Therefore, $M = 0$ is not an equilibrium of the system.

Moreover, for $M(t)$ sufficiently small, we have

$$\frac{dM}{dt} \geq b_0 K_0 - \mu M(t). \quad (33)$$

Choosing $\varepsilon > 0$ such that $\varepsilon < \frac{b_0 K_0}{\mu}$, it follows that

$$\frac{dM}{dt} > 0 \quad \text{whenever } 0 < M(t) < \varepsilon.$$

Thus, any trajectory entering a neighborhood of zero is driven away from it.

Consequently, $M(t)$ cannot converge to zero as $t \rightarrow \infty$, and the extinction state is not asymptotically stable. $\square$

3.2. Basic Reproduction Number ($\mathcal{R}_0$)

Following the next-generation matrix method of van den Driessche and Watmough [33], we order the infected compartments as $X_I = (E_H, I_H, E_M, I_M)^T$ and denote by $\mathcal{F}_i(x)$ the rate of appearance of new infections in compartment i and by $\mathcal{V}_i(x) = \mathcal{V}_i^-(x) - \mathcal{V}_i^+(x)$ the net rate of transfer of individuals into and out of compartment i . The disease transmission model consists of non-negative initial conditions together with the system $\dot{x} = f(x) = \mathcal{F}(x) - \mathcal{V}(x)$, where each component is assumed twice continuously differentiable.

Disease-free equilibrium: The disease-free equilibrium (DFE) is

$$\begin{aligned} x_0 &= (S_H^*, E_H^*, I_H^*, R_H^*, S_M^*, E_M^*, I_M^*)^T \\ &= (N, 0, 0, 0, M^*, 0, 0)^T. \end{aligned} \quad (34)$$

The equilibrium mosquito abundance M^* is obtained from the mosquito subsystem

$$\frac{dM}{dt} = \Lambda(t) - \mu M = b(R, T) K(t) \left(1 - \frac{M}{K(t)}\right) - \mu M = 0, \quad (35)$$

which yields

$$M^* = \frac{\bar{b} \bar{K}}{\bar{b} + \mu}, \quad \Lambda^* := \Lambda \Big|_{M=M^*} = \mu M^* = \frac{\mu \bar{b} \bar{K}}{\bar{b} + \mu}, \quad (36)$$

where $\bar{b} = b(\bar{R}, \bar{T})$ and $\bar{K} = K(t)$ are the environmental drivers evaluated at a fixed reference state, that being their study-period means, so that $\mathcal{R}_0$ is a function of that reference climate-habitat state. Note that Λ^* is *not* an arbitrary value: it is the recruitment rate that sustains the equilibrium vector population M^* , and it inherits the environmental dependence of $b(R, T)$ and $K(t)$.

Next-generation matrices: The new-infection and transfer matrices, evaluated at the DFE, are

$$F = \begin{pmatrix} 0 & 0 & 0 & \frac{ab_2 S_H^*}{N} \\ 0 & 0 & 0 & 0 \\ 0 & \frac{ab_1 S_M^*}{N} & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad (37)$$

$$V = \begin{pmatrix} b_{3H} & 0 & 0 & 0 \\ -b_{3H} & \gamma & 0 & 0 \\ 0 & 0 & b_{3M}\ell + \mu & 0 \\ 0 & 0 & -b_{3M}\ell & \mu \end{pmatrix},$$

where we used $S_H^* = N$ and $S_M^* = M^*$ at the DFE. The spectral radius of FV^{-1} gives

$$R_0 = \rho(FV^{-1}) = \sqrt{\frac{a^2 b_1 b_2 \Lambda^* b_{3M} \ell}{N \gamma (b_{3M} \ell + \mu) \mu^2}}, \quad (38)$$

which can be written in the classical two-factor host-vector form

$$R_0^H = \frac{ab_2}{\gamma}; \quad R_0^M = \frac{ab_1 \Lambda^*}{N \mu^2} \cdot \frac{b_{3M} \ell}{b_{3M} \ell + \mu} \quad (39)$$

$$\Rightarrow R_0 = \sqrt{R_0^H \cdot R_0^M},$$

where R_0^H is the expected number of humans infected by a single infectious mosquito, and R_0^M is the expected number of mosquitoes that survive the latent period and become infectious, generated by a single infectious human. The geometric mean arises from the two-generation host-vector transmission cycle: the infection must complete a full human→mosquito→human loop, and the block structure of the next-generation matrix decouples into these two factors (see Appendix A.2 for the explicit multiplication).

We note that R_0 is independent of both the intrinsic incubation period τ_H and the immunity waning rate ω : the former because the model has no human disease-induced mortality (every exposed individual eventually becomes infectious, so the latent period affects only the timing of secondary infections), and the latter because no individuals are immune at the disease-free equilibrium, so waning is irrelevant to the linearized invasion dynamics.

3.3. Sensitivity analysis methodology

The reproduction number in Eq. (39) depends on the reference habitat state through the mosquito recruitment $\Lambda^* = \mu M^*$, which inherits the environmental dependence of the recruitment function $b(\bar{R}, \bar{T})$ and of the carrying capacity $\bar{K}$ (Eq. 36). Consequently, R_0 is a function of the calibrated parameters and of the environmental drivers. Because these quantities enter nonlinearly and are mutually coupled, the relative contribution of each to the invasion threshold is not apparent from the analytical expression alone, and must be quantified numerically.

4. Results

4.1. Recalibration of the temperature-dependent biting rate

The temperature-dependent biting rate was recalibrated using field estimates of *Anopheles darlingi* feeding behavior, as described in Section 2.6. The recalibrated function has a minimum biting temperature threshold of $T' = 16^\circ\text{C}$ and a thermal scaling factor $D_1 = 24.8^\circ\text{C days}$. Compared with the original Parham & Michael parameterization, the recalibrated function better reflects the temperature range observed in the Amazon and produces higher biting rates during the transmission season in Manaus (Fig. 2). This recalibrated biting rate is used in all subsequent simulations and in the weekly transmission regressions presented in Section 4.4.

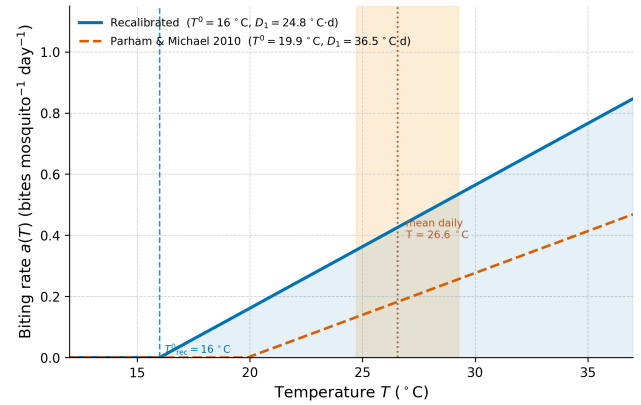


Figure 2: Comparison between the recalibrated temperature-dependent biting rate $a(T)$ used in this work ($T' = 16^\circ\text{C}$, $D_1 = 24.8^\circ\text{C days}$) and the original Parham & Michael (2010) parameterization [11] ($T' = 19.9^\circ\text{C}$, $D_1 = 36.5^\circ\text{C days}$). The shaded band indicates the range of daily mean temperatures observed in Manaus between 2017 and 2023 (5th-95th percentiles), and the dashed vertical line marks the recalibrated biting threshold $T' = 16^\circ\text{C}$.

4.2. Model calibration reproduces the observed transmission dynamics

The SEIRS-SEI model was calibrated year by year over 2017-2023 following the sequential topic scheme of Section 2.9, and the recovered parameters are summarized in Table 2. Figure 3 shows the fitted infectious compartment I_H against the observed active-case count alongside the weekly free-beta fit for 2017, which exhibited the largest relative improvement of the weekly model over the constant-coefficient formulation (Section 4.3). The calibrated parameters remain within their physiologically prescribed bounds found in Table 1, with the humidity response (H_0, k, ϕ) and the human recovery parameters (τ_H, γ, ω) showing limited cross-year variation, while the force-of-infection coefficients b_1 and b_2 consistently approach their upper and lower bounds, respectively (Table 2).

Table 2

Year-by-year calibrated parameters and weighted mean squared error (wMSE) of the SEIRS-SEI model, 2017-2023.

Year	H_0	k	φ	τ_H	γ	ω	b_1	b_2	M'	wMSE
2017	37.58	0.705	0.215	19.81	0.0071	0.0855	0.259	0.035	50,723	85,036
2018	20.36	0.046	0.490	20.00	0.0073	0.0961	0.487	0.022	51,245	64,959
2019	22.13	0.053	0.493	19.96	0.0072	0.0744	0.488	0.022	50,474	41,436
2020	20.13	0.039	0.500	19.98	0.0067	0.0422	0.500	0.019	50,024	23,541
2021	20.02	0.043	0.493	19.99	0.0067	0.0773	0.500	0.019	50,015	39,449
2022	20.02	0.043	0.493	19.77	0.0099	0.0312	0.500	0.019	50,832	25,046
2023	20.14	0.207	0.431	19.86	0.0081	0.0310	0.485	0.010	53,616	18,360

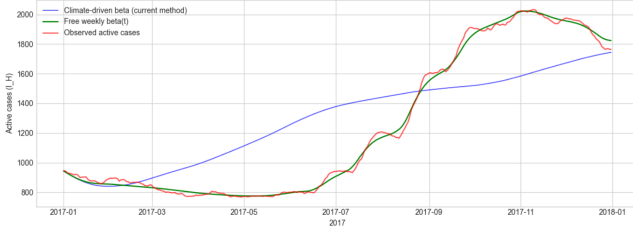
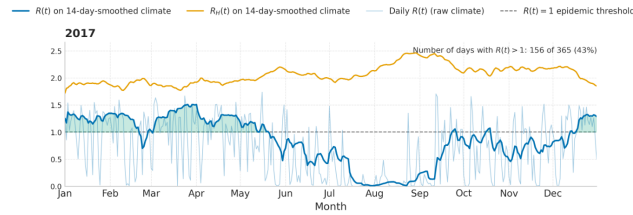
**Figure 3:** Comparison between the observed active cases and the model estimation of the infectious human compartment I_H for 2017.**Figure 4:** Seasonal variation of the coupled and human basic reproduction number $R(t)$ for the calibrated model in 2017.

Figure 4 shows the seasonal variation of both human and coupled $R(t)$ during 2017, obtained by substituting the calibrated parameters and the observed drivers into Eq. (39). The reproduction number follows the annual transmission cycle, exceeding unity during the wet season and dropping below the epidemic threshold in the months of lowest precipitation. Because $\Lambda^* = \mu M^*$ inherits the biting rate and carrying capacity as functions of the environmental drivers, this curve makes explicit the dependence of the invasion threshold on the climate-habitat state highlighted in Section 3.2, and motivates the week-specific regression of the transmission probabilities pursued in Section 4.4.

The biological survival rates A and B (which carry the temperature dependence of the vector) dominate R_0 , followed by the biting rate a and the mosquito mortality μ . These are precisely the vector vital rates coupled to the environmental drivers, confirming that the climate-habitat state modulates the invasion threshold as established in Section 3.2. The transmission probabilities b_1 and b_2 and the recruitment-related terms Λ , M' and H_0 display intermediate sensitivity, while the vector competence ℓ contributes weakly.

Table 3

Weighted mean squared error (wMSE) of the climate-driven constant-coefficient model versus the free weekly model, per year.

Year	Climate-driven	Weekly free	Reduction factor
2017	85,036	618	137.6×
2018	64,959	2,553	25.4×
2019	41,436	561	73.9×
2020	23,541	359	65.6×
2021	39,449	1,552	25.4×
2022	25,046	1,711	14.6×
2023	18,360	7,598	2.4×

Critically, the PRCCs for the average duration of immunity ω and the latent period of infection τ_H are indistinguishable from zero. This is the numerical counterpart of the analytical factorisation in Eq. (39), in which these two terms cancel exactly and therefore do not enter R_0 . The agreement between the algebraic structure and the empirical sensitivity provides independent support for the two-factor formulation derived in Section 3.2.

4.3. The constant-coefficient climate model underfits the observed dynamics

To assess how much of the seasonal variation in transmission is captured by the mechanistic, climate-driven formulation, we freed the effective transmission coefficients, allowing them to vary on a weekly basis (Section 2.10), and compared the resulting weekly model with the constant-coefficient climate-driven model. Releasing the coefficients reduced the weighted mean squared error in every year (Table 3), most notably in 2017, where the wMSE fell from 8.5×10^4 to 6.2×10^2 (a ~ 137 -fold reduction).

The residual between the free weekly $\beta_{h,w}, \beta_{m,w}$ and the climate-driven $a(T)b_1, a(T)b_2$ indicates periods in which the temperature-driven formulation fails to capture the observed transmission structure. This motivates extending the per-bite transmission probabilities to the wider set of environmental drivers introduced in Section 1.

Table 4

Year-by-year within-year random forest performance for b_1 and b_2 .

	Year	R^2	MAE	Bias	Top Driver
b_1	2017	0.563	0.0594	+0.0021	T
	2018	0.580	0.0651	-0.0005	T
	2019	0.739	0.0475	+0.0011	F_c
	2020	0.671	0.0626	+0.0021	R
	2021	0.705	0.0603	-0.0009	F_t
	2022	0.810	0.0503	+0.0010	T
	2023	0.724	0.0107	+0.0006	F_t
b_2	2017	0.531	0.0182	+0.0011	R
	2018	0.825	0.0103	+0.0005	T
	2019	0.712	0.0120	-0.0002	H
	2020	0.915	0.0105	-0.0006	F_t
	2021	0.712	0.0122	+0.0001	R
	2022	0.905	0.0103	-0.0002	F_t
	2023	0.889	0.0021	+0.0001	F_c

4.4. Environmental regression of the transmission probabilities b_1 and b_2

Following the framework described in Section 1, the freed weekly per-bite transmission probabilities were regressed against the environmental predictors T , R , H , D_f , F_c , F_t , representing temperature, precipitation, humidity, deforestation, fire counts and fire trends. A separate random forest model was fitted for each year.

Within each year the model explains a large part of the variance of both transmission probabilities, with prediction bias was close to zero in all years ($|\text{bias}| \lesssim 0.003$ in every year; Table 4).

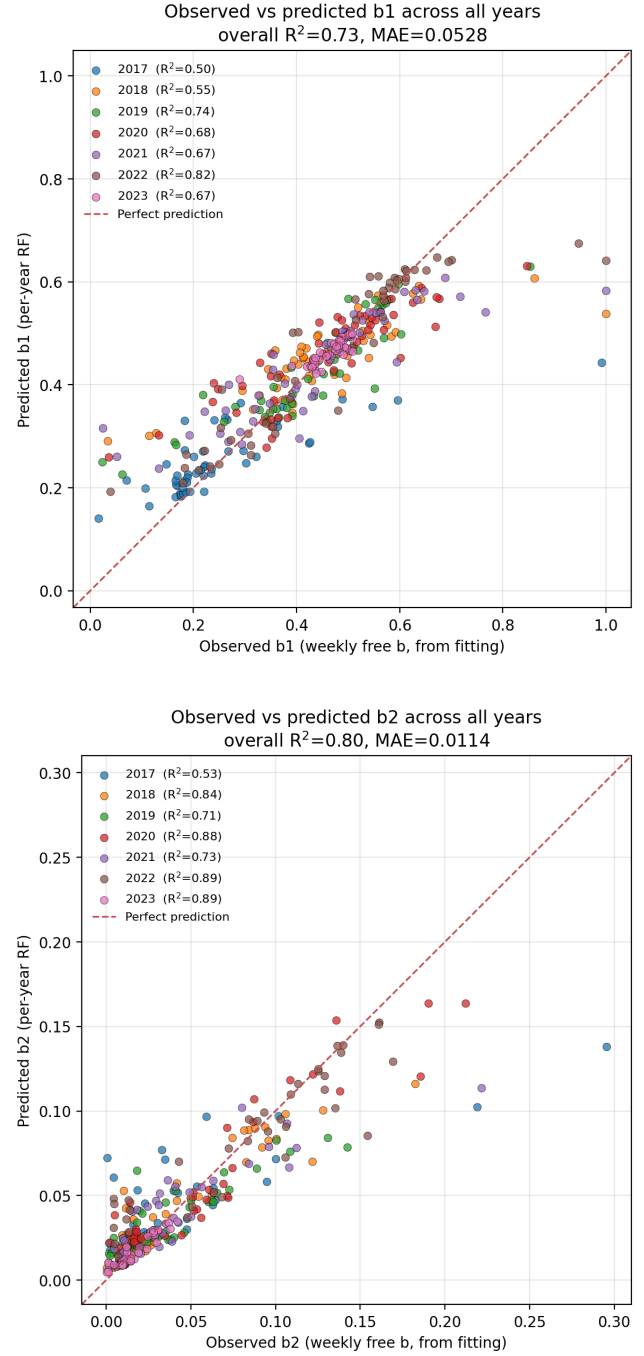


Figure 5: Scatter plot of the estimated vs predicted weekly-free proportion bites from infected humans to susceptible mosquitoes (b_1) and from infected mosquitoes to susceptible humans (b_2) that result in infection.

The importance of the predictors, however, differs between the two probabilities, as can be seen in Table 5. Based on this, it can be seen that deforestation is consistently the weakest predictor for both b_1 and b_2 , indicating that its effect is not captured at the intra-annual timescale resolved here. More strikingly, precipitation and fire counts occupy opposite ends of the two rankings: precipitation is the leading predictor of b_1 but only second-lowest for b_2 ,

Table 5Relative importance of environmental predictors for b_1 and b_2 .

	b_1	b_2
Temperature (T)	20.1%	16.9%
Precipitation (R)	20.3%	16.3%
Humidity (H)	19.4%	17.1%
Deforestation (D_f)	7.4%	11.8%
Fire Counts (F_c)	14.9%	21.3%
Fire Trends (F_t)	17.9%	16.6%

whereas fire counts lead for b_2 but rank second-lowest for b_1 . Temperature and humidity, by contrast, consistently hold the second and third positions for both targets, merely exchanging order: temperature then humidity for b_1 , humidity then temperature for b_2 .

A leave-one-year-out cross-validation of the same forests qualifies this result. Out-of-sample performance is markedly lower, with overall $R^2 \approx 0.17$ for b_1 and ≈ 0.19 for b_2 , in contrast to the strong within-year fits. This contrast confirms that the environmental drivers are strongly year-specific: the within-year seasonal trajectory of T, R, H, D_f, F_c, F_t shapes each year's transmission signal, whereas a fixed, year-independent link between the environment and b_1, b_2 is not supported by the data.

The shape of these responses is illustrated in Figs. 6 and 7, which shows the partial dependence of b_1 and b_2 on each predictor for the best fitted year for each transmission probability.

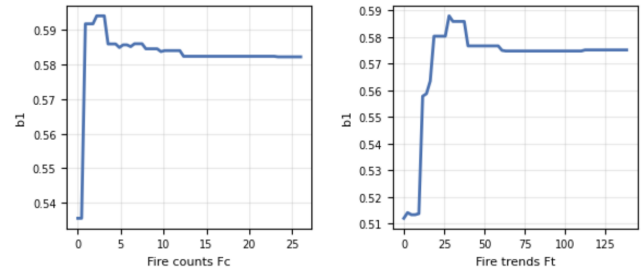
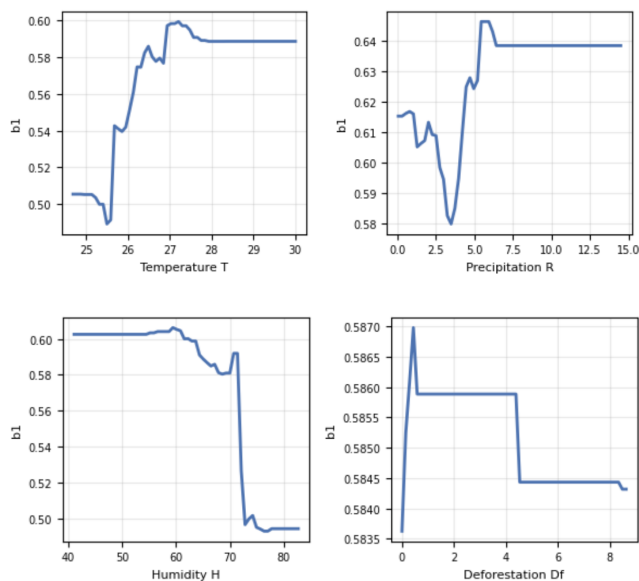


Figure 6: Partial dependence of the weekly per-bite transmission probability b_1 on each environmental predictor in 2022.

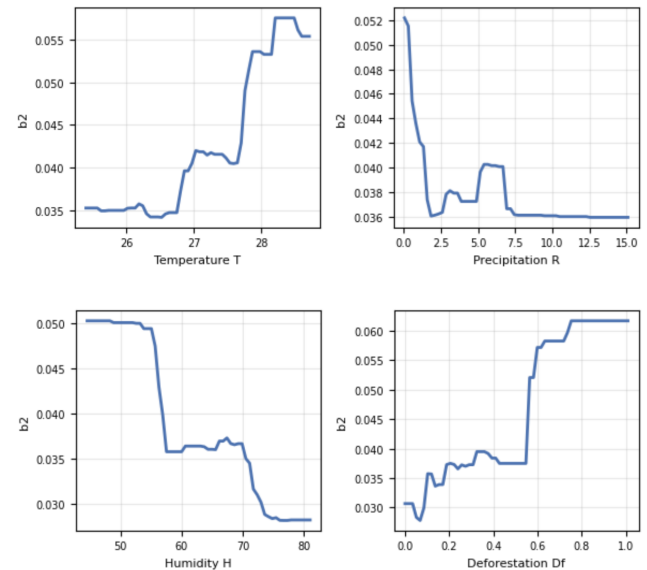


Figure 7: Partial dependence of the weekly per-bite transmission probability b_2 on each environmental predictor in 2020.

Columns correspond to temperature T , precipitation R , humidity H , deforestation D_f , fire counts F_c and fire trends F_t . Curves give the mean predicted probability as each predictor is varied over its observed range while the remaining variables are held at their marginal mean.

5. Discussion

5.1. Reproduction of the original Parham & Michael model as a negative baseline

As a diagnostic for the methodological choices adopted below, the original SIR-SEI formulation of Parham and Michael (2010) [11] was implemented with all parameter values reported in the original study, while population sizes and initial epidemiological conditions were estimated from the available rural Manaus data. Under this parameterization the infected human population declines monotonically and the mosquito population rapidly collapses to extinction, indicating that mosquito recruitment is insufficient to replace mortality. Because transmission depends on the persistence of infectious vectors, the collapse of the vector population eliminates infection in both host and vector populations. Although the delay formulation preserved positivity, it fails to reproduce the endemic behavior expected for malaria in the Amazon, confirming the need to recalibrate the environmental and transmission parameters as described in Section 2.6 and Section 2.9. This can be seen in the Supplementary Material.

5.2. The model sustains endemic transmission where the baseline does not

The transition from the delayed SIR-SEI model of Parham and Michael to the ODE-based SEIRS-SEI model was not simply a numerical convenience. As shown in Section 2.3, the original parameterization leads to the extinction of both infected humans and mosquitoes, since the linear mosquito recruitment term is not sufficient to compensate for mosquito mortality under the environmental conditions of Manaus. In the modified model, the explicit extrinsic incubation delay is replaced by the transition rate $b_{3M}\ell$, while mosquito recruitment is made density-dependent. These changes allow the model to sustain the mosquito population and reproduce endemic transmission, as established in Proposition 1. Thus, the modifications change the qualitative behavior of the system and are important for obtaining a model that is consistent with the observed persistence of malaria transmission in Manaus.

5.3. Two-factor $\mathcal{R}_0$: the transmission probabilities drive the invasion threshold

The next-generation analysis (Section 3.2) yields a reproduction number that factorises into a human-to-mosquito and a mosquito-to-human component,

$$\mathcal{R}_0 = \sqrt{\mathcal{R}_0^H \mathcal{R}_0^M},$$

a form that makes the two-step transmission cycle explicit and facilitates interpretation. The global sensitivity analysis (Section 3.3) confirms that the transmission probabilities b_1 and b_2 dominate the variation of the invasion threshold, and that the intrinsic incubation period τ_H and the immunity waning rate ω are effectively irrelevant to it. This corroborates the analytical result: in the absence of disease-induced mortality, every exposed individual

eventually becomes infectious, so the latent period only shifts the timing of secondary infections, while no individual is immune at the disease-free equilibrium, so waning does not enter the linearised invasion dynamics.

Because Λ^* and M^* inherit the biting rate and carrying capacity as functions of precipitation, temperature and habitat, the reproduction number is itself a function of the environmental state. The seasonal $\mathcal{R}_0$ curve of Figure 4 makes this dependence explicit for 2017: it exceeds unity during the wet season and falls below the epidemic threshold in the driest months, confirming that the invasion risk is not constant throughout the year.

5.4. Constant coefficients underfit: the transmission probabilities are seasonally heterogeneous

The constant-coefficient model consistently underestimates the observed dynamics, as the reduction in the weighted mean squared error between the free-weekly and constant formulations demonstrates (Table 3). The improvement ranges from a ~ 137 -fold reduction in 2017 to a ~ 2.4 -fold reduction in 2023, indicating that the constant-coefficient formulation already captures most of the signal in the latter year, while the transmission probabilities vary more strongly across weeks in the former. This heterogeneity is the empirical signature of the environmental dependence of transmission, and it motivates the week-specific regression of $b_{1,w}$ and $b_{2,w}$ in Section 4.4.

5.5. Random forest: strong within years, weak in-between

The random forest regression achieves a good within-season fit ($R^2 \approx 0.6$ - 0.9) but a markedly poorer between-season fit ($R^2 \approx 0.17$ - 0.19). This asymmetry is informative rather than a failure of the model. Within a single year, the drivers that vary on the weekly scale (precipitation, temperature, humidity, and fire) largely determine the week-to-week fluctuations of the transmission probabilities. Across years, however, the differences are not captured by these same variables, suggesting that inter-annual variation is governed either by factors that are constant within a year in our dataset or by an underlying year-specific environmental state that is not fully represented by the contemporaneous drivers. The poor performance points to a structural limitation of the driver set rather than to an insufficiency of the statistical method.

5.6. Deforestation is a weak within-season predictor

Contrary to the expectation that land-use change drives transmission, deforestation ranked as one of the weakest predictors at the within-season scale. This does not support a negligible role for deforestation in general, suggesting in fact that within a single year, the deforestation signal is nearly constant and therefore carries little information about week-to-week variation in the transmission probabilities.

At the between-season scale, where land-use change could plausibly matter, the random forest performs poorly, precisely because the within-year variability of deforestation does not discriminate among years. This is consistent with the hypothesis that habitat modification contributes to long-term changes in transmission intensity, even if it is not captured by a short within-season regression. The fire term, by contrast, contributes meaningfully to the variation of the mosquito-to-human probability b_2 , consistent with the seasonal burning cycle in the region.

5.7. Calibration reaches the bounds

The calibrated transmission probabilities b_1 and b_2 reach the bounds of their respective ranges (Table 1), with b_1 reaching its upper bound in 2020–2022 while b_2 reaches its lower limit over the same period. This suggests that the environmental conditions affecting transmission in Manaus may not be fully captured by the original literature-based parameter ranges. It also indicates that the density-dependent mosquito recruitment and the recalibrated biting rate play an important role in reproducing the observed dynamics. At the same time, the parameters reaching their bounds highlight a limitation of the fitting procedure, suggesting that a wider parameter range or a different functional form may be needed to better capture the environmental effects on transmission.

5.8. Limitations

Several limitations should be acknowledged. First, as mentioned in Section 2.2, the ratio of rural to total population had to be recalculated, given that the ratios from the Trajetórias Project [12] for the years of 2006 and 2017 both used the estimated rural and urban populations from the 2010 census, which conceal the decline of the rural population from 2000 to 2010. In addition to that, other inconsistencies were identified in the dataset: in particular, incidence rates for the period 2015–2019 had been calculated using population estimates corresponding to 2004–2008, resulting in a temporal mismatch. These values were recalculated using population estimates consistent with the appropriate time interval. Second, the environmental drivers are treated as deterministic and observed; unobserved covariates, such as vector control interventions or local mobility, are not represented and may contribute to the between-season degradation of the random forest. Third, the model is fitted to reported case counts, which may be subject to reporting bias and under-reporting, particularly in the lowest-transmission months.

6. Conclusion

Supplementary Material

All files of data collection and treatment, as well as results obtained throughout the development of this work, are available at https://github.com/RaphaLevy/Environmental_Drivers_of_Malaria.

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A. Appendix

A.1. Tables

Table A.1

List of model parameters.

Parameter	Definition	Formulation
$p(T)$	Daily mosquito survival rate dependent on temperature	$e^{(-1/(AT^2+BT+C))}$
$p(H)$	Daily mosquito survival rate dependent on humidity ¹	$\frac{\varphi e^{-k(H-H_0)+1}}{e^{-k(H-H_0)+1}+1}$
$p(T, H)$	Daily mosquito survival rate	$p(T)p(H)$
$p_L(R)$	Probability of larval survival dependent on rainfall ²	$(4p_{ML}/R_L^2)R(R_L - R)$
$p_L(T)$	Probability of larval survival dependent on temperature	$e^{-(c_1T+c_2)}$
$p_L(R, T)$	Probability of larval survival	$p_L(R)p_L(T)$
$\mu(T, H)$	Mosquito mortality rate per capita (days ⁻¹)	$-\log(p(T, H))$
$a(T)$	Biting rate (days ⁻¹)	$(T - T')/D_1$
$\tau_M(T)$	Duration of sporozoite cycle (days)	$DD/(T - T_{\min})$
$\tau_L(T)$	Duration of larval development phase (days)	$1/(c_1T + c_2)$
$\ell(\tau_M)(T, H)$	Probability of mosquito survival during sporozoite cycle (days ⁻¹)	$p(T, H)^{\tau_M(T)}$
$b(R, T)$	Mosquito birth rate (days ⁻¹)	$B_E p_E(R) p_L(R, T) p_P(R) / (\tau_E + \tau_L(T) + \tau_P)$
$M(t)$	Total population of mosquitoes	$S_M(t) + E_M(t) + I_M(t)$
$N(t)$	Total population of humans	$S_H(t) + E_H(t) + I_H(t) + R_H(t)$

¹ $p(H)$ was added in order to take into consideration the deleterious effects of humidity. Unlike for T and R , which use the daily average value of temperature and precipitation, respectively, H uses the daily minimal value of humidity.

² $p_E(R)$ and $p_P(R)$ have an analogous expression to $p_L(R)$, substituting p_{ML} for p_{ME} and p_{MP} respectively.

Table A.2

List of model variables.

Symbol	Definition	Units
b_1	Proportion of bites from susceptible mosquitoes on infected humans that result in infection: 0.04	Dimensionless
b_2	Proportion of bites from infected mosquitoes on susceptible humans that result in infection: 0.09	Dimensionless
γ	1/Average duration of infectiousness in humans: 1/120	days ⁻¹
ω	1/Average duration of protective immunity after infection: 1/270	days ⁻¹
B_E	Number of eggs laid per adult per oviposition: 200	Dimensionless
p_{ME}	Maximum probability of egg survival: 0.9	Dimensionless
p_{ML}	Maximum probability of larval survival: 0.25	Dimensionless
p_{MP}	Maximum probability of pupal survival: 0.75	Dimensionless
τ_E	Duration of the egg development phase: 1	days
τ_P	Duration of the pupal development phase: 1	days
τ_H	Latent period of infection within humans: 10	days
R_L	Rainfall threshold until breeding sites are eliminated, removing immature individuals: 32.26 ³	mm
$T_{\min}$	Minimum temperature, below which there is no development of the parasite: 14.5	°C
$T_{\max}$	Maximum temperature, above which there is no development of the parasite: 29.8 [35]	°C
T_{opt}	Optimal temperature for development of the parasite: 23.5 [35]	°C
$T_{\min A.}/T'$	Minimum temperature, below which there is no development of the vector: 16 ⁴ [24, 36]	°C
$T_{\max A.}$	Maximum temperature, above which there is no development of the vector: 34 [24, 36, 37]	°C
$T_{\text{op } A.}$	Optimal temperature for transmission: 25 [24]	°C
DD	Degree-days for parasite development. "Sum of heat" for maturation: 105 [35, 38]	°C days
A	Empirical sensitivity parameter: -0.03	(°C ² days) ⁻¹
B	Empirical sensitivity parameter: 1.31	(°C days) ⁻¹
C	Empirical sensitivity parameter: -4.4	days ⁻¹
D_1	Temperature scaling factor of the biting rate: 24.8	°C days
c_1	Empirical sensitivity parameter: 0.00554	(°C days) ⁻¹
c_2	Empirical sensitivity parameter: -0.06737	days ⁻¹
b_{3M}	Infection rate in exposed mosquitoes ($1/\tau_M(T)$)	days ⁻¹
b_{3H}	Infection rate in exposed humans ($1/\tau_H$): 1/10	days ⁻¹
H_0	Critical humidity threshold for 50% survival transition: 58	%
φ	Baseline survival probability at extreme low humidity: 0.05	Dimensionless
k	Steepness of the humidity-survival sigmoid response: 0.25	% ⁻¹

³ The reference rainfall level R_L for the Amazon was approximated based on reported associations between monthly precipitation and malaria incidence [39]. In particular, precipitation levels in the range of 917 to 1019 mm/month were associated with a decrease in malaria cases. The midpoint of this interval (968 mm/month) was used as a representative value and converted to a daily rate by dividing by 30 days, yielding $R_L \approx 32.3$ mm/day. This value is therefore interpreted as the threshold scale for high rainfall conditions in the model.

⁴ In Parham and Michael [11], this parameter is denoted by T_1 . Here, we adopt a different notation to avoid confusion with the mean temperature in the absence of seasonality, also denoted by T_1 in the original paper. Additionally, instead of defining an independent lower thermal threshold for the biting rate, we assume that biting activity becomes negligible below the minimum temperature required for vector development.

A.2. Derivation of the basic reproduction number

This appendix details the next-generation matrix (NGM) computation summarised in Section 3.2. The infected compartments are ordered as (E_H, I_H, E_M, I_M) , so $m = 4$, and all parameters are as defined in Section 3.

Disease-free states and the NGM formalism: Following van den Driessche and Watmough [33], let X_s be the set of all disease-free states:

$$X_s = \{x \geq 0 \mid x_i = 0, i = 1, \dots, m\},$$

where $X = (x_1, \dots, x_n)^\top$, $x_i \geq 0$ is the number of individuals in compartment i , and every function is assumed to be continuously differentiable at least twice in each variable (C^2).

Rearranging the model equations so that the first m components are the infected compartments, let $\mathcal{F}_i(x)$ be the rate of appearance of new infections in compartment i , $\mathcal{V}_i^+(x)$ the rate of transfer into compartment i by all other means, and $\mathcal{V}_i^-(x)$ the rate of transfer out of compartment i . The model can then be written as

$$\dot{x}_i = f_i(x) = \mathcal{F}_i(x) - \mathcal{V}_i(x), \quad i = 1, \dots, n,$$

with $\mathcal{V}_i(x) = \mathcal{V}_i^-(x) - \mathcal{V}_i^+(x)$. The next-generation matrices are the Jacobians of these terms evaluated at the disease-free equilibrium x_0 :

$$F = \left[\frac{\partial \mathcal{F}_i(x_0)}{\partial x_j} \right], \quad V = \left[\frac{\partial \mathcal{V}_i(x_0)}{\partial x_j} \right], \quad 1 \leq i, j \leq m,$$

so that $\mathcal{R}_0 = \rho(FV^{-1})$.

Disease-free equilibrium and mosquito demographic equilibrium: At the DFE the human population is fully susceptible, $S_H^* = N$, and all infected compartments vanish. The mosquito population rests at the demographic equilibrium M^* of the density-dependent recruitment dynamics (11). Because the climate drivers are time-dependent, we evaluate them at their fixed reference state (the study-period means $\bar{T}$, $\bar{R}$), so that the recruitment rate and the carrying capacity become constants:

$$\bar{b} = b(\bar{R}, \bar{T}), \quad \bar{K} = K(t) \Big|_{\text{DFE}} = M' f_T(\bar{T}) f_R(\bar{R}),$$

and

$$\frac{dM}{dt} = \bar{b} \bar{K} \left(1 - \frac{M}{\bar{K}} \right) - \mu M$$

Imposing $\Lambda(M^*) = \mu M^*$ gives

$$\bar{b} \bar{K} \left(1 - \frac{M^*}{\bar{K}} \right) = \mu M^* \implies M^* = \frac{\bar{b} \bar{K}}{\bar{b} + \mu},$$

and hence the recruitment at the DFE is

$$\Lambda^* = \mu M^* = \frac{\mu \bar{b} \bar{K}}{\bar{b} + \mu}$$

By Proposition 1 (Section 3.1), $M^* > 0$ is the unique relevant equilibrium whenever $\bar{b}, \bar{K} > 0$. Therefore, the disease-free equilibrium will be

$$x_0 = (N, 0, 0, 0, M^*, 0, 0)^\top. \quad (40)$$

Verification of $\mathcal{R}_0$ by explicit multiplication: From (37), the transfer matrix is block-diagonal, $V = \text{diag}(V_H, V_M)$, with

$$V_H = \begin{pmatrix} b_{3H} & 0 \\ -b_{3H} & \gamma \end{pmatrix}, \quad V_M = \begin{pmatrix} b_{3M}\ell + \mu & 0 \\ -b_{3M}\ell & \mu \end{pmatrix},$$

so that

$$V^{-1} = \text{diag}(V_H^{-1}, V_M^{-1}),$$

$$V_H^{-1} = \begin{pmatrix} \frac{1}{b_{3H}} & 0 \\ \frac{1}{\gamma} & \frac{1}{\gamma} \end{pmatrix},$$

$$V_M^{-1} = \begin{pmatrix} \frac{1}{b_{3M}\ell + \mu} & 0 \\ \frac{1}{\mu(b_{3M}\ell + \mu)} & \frac{1}{\mu} \end{pmatrix}$$

Carrying out the block multiplication FV^{-1} yields the hollow form

$$FV^{-1} = \begin{pmatrix} 0 & P \\ Q & 0 \end{pmatrix},$$

$$P = \begin{pmatrix} \frac{ab_2 S_H^* b_{3M}\ell}{N\mu(b_{3M}\ell + \mu)} & \frac{ab_2 S_H^*}{N\mu} \\ 0 & 0 \end{pmatrix},$$

$$Q = \begin{pmatrix} \frac{ab_1 S_M^*}{N\gamma} & \frac{ab_1 S_M^*}{N\gamma} \\ 0 & 0 \end{pmatrix}$$

Both P and Q have rank one, so PQ has a single nonzero eigenvalue $\lambda = \frac{a^2 b_1 b_2 S_H^* S_M^* b_{3M}\ell}{N^2 \mu \gamma (b_{3M}\ell + \mu)}$, and the spectrum of FV^{-1} is

$$\sigma(FV^{-1}) = \{0, 0, \pm\sqrt{\lambda}\} = \{0, 0, \pm\sqrt{\mathcal{R}_0^H \mathcal{R}_0^M}\}$$

Substituting $S_H^* = N$ and $S_M^* = M^* = \Lambda^*/\mu$ into λ and taking the square root reproduces (38).

Spectral computation of $\rho(FV^{-1})$: The matrix V is block-diagonal in the human-mosquito partition, with blocks V_H and V_M , so

$$V^{-1} = \begin{pmatrix} \frac{1}{b_{3H}} & 0 & 0 & 0 \\ \frac{1}{\gamma} & \frac{1}{\gamma} & 0 & 0 \\ 0 & 0 & \frac{1}{\mu + b_{3M}\ell} & 0 \\ 0 & 0 & \frac{b_{3M}\ell}{(\mu + b_{3M}\ell)\mu} & \frac{1}{\mu} \end{pmatrix}$$

Multiplying F by V^{-1} we obtain

$$FV^{-1} = \begin{pmatrix} 0 & 0 & \frac{ab_2 b_{3M}\ell}{(\mu + b_{3M}\ell)\mu} & \frac{ab_2}{\mu} \\ 0 & 0 & 0 & 0 \\ \frac{ab_1 M^*}{N\gamma} & \frac{ab_1 M^*}{N\gamma} & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

The second and fourth rows are zero, so the spectral problem reduces to the 2×2 block coupling the human and mosquito exposed classes,

$$[FV^{-1}]_{\{1,3\}} = \begin{pmatrix} 0 & \frac{ab_2b_{3M}\ell}{(\mu + b_{3M}\ell)\mu} \\ \frac{ab_1M^*}{N\gamma} & 0 \end{pmatrix},$$

whose characteristic equation is

$$\lambda^2 = \frac{a^2b_1b_2b_{3M}\ell M^*}{N\gamma(\mu + b_{3M}\ell)\mu}$$

The eigenvalues of FV^{-1} are $\lambda \in \{0, 0, \pm\sqrt{\mathcal{R}_H\mathcal{R}_M}\}$, where

$$\mathcal{R}_H = \frac{ab_2}{\gamma}, \quad \mathcal{R}_M = \frac{ab_1\Lambda^*}{N\mu} \frac{b_{3M}\ell}{(b_{3M}\ell + \mu)\mu},$$

so that $\mathcal{R}_0 = \rho(FV^{-1}) = \sqrt{\mathcal{R}_H\mathcal{R}_M}$. Finally, substituting $\Lambda^* = \mu M^*$ recovers the expression in Section 3.2:

$$\mathcal{R}_0 = \sqrt{\frac{a^2b_1b_2\Lambda^*b_{3M}\ell}{N\gamma(b_{3M}\ell + \mu)\mu^2}} \quad (41)$$